iGEM 2026: Model and Software Strategy

Dry Lab computational workstream. What we are building, who owns what, and by when.

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Read this first

This document exists so nobody on the Dry Lab side has to ask "what am I supposed to be doing". Every task below has a named owner, a concrete deliverable, and a rough due window. If your name is not on a task, it is not your job right now. If something below is unclear, that is a bug in this document, flag it and we fix the document, not guess.

We are going for two iGEM special prizes: **Best Model** and **Best Software Tool**. Both are built from one underlying idea: turning the protease cleavage event in our construct into a predictable, quantitative signal. The model predicts how that signal behaves. The software tool (later, not now) lets other people use that prediction without understanding the maths.

Active now: the kinetics model (**Archer and Aniu**) and the structural feasibility scorer (**Aaran**). **Active now, different deliverable:** the wiki build (**Orlando**). **Not started yet, on purpose:** the web and command line software tool (**Orlando and Aaran**, later). We talk about it below so the spec exists when we are ready, but no code gets written for it yet.

What actually wins these two prizes

iGEM judges score against a fixed rubric. Below is the rubric, word for word in spirit, taken from the official judging materials. Everything else in this document is just a plan to satisfy these lines.

Best Model. Three questions.

1. Did the model help the team understand, predict, or design a part, device, or system. Not explain after the fact. Predict, before or alongside the wet lab.
2. Did the team use experimental or other relevant data to develop, train, or validate the model.
3. Does the modelling approach provide a good example for other teams to reuse.

Source: competition.igem.org/judging/awards/special and the iGEM Judging Handbook.

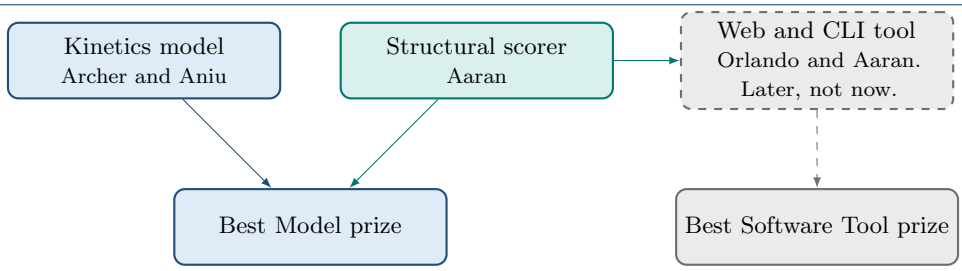
Best Software Tool. Six questions.

1. Does it use existing synthetic biology data standards where relevant.
2. Was it validated by real experimental work, not just simulated against itself.
3. Can it be useful to other iGEM teams, not just us.
4. Does it integrate with other tools, packages, or APIs.
5. Is it actually usable by someone who is not us.
6. Is it documented well enough for a future team to extend it.

Source: the judging criteria published on a 2025 award winning team's software page, 2025.igem.wiki/lambert-ga/software.

Hard rule. Software submitted for this prize must be hosted on iGEM's own GitLab software tools repository before the GitLab freeze. A tool that only lives on someone's laptop wins nothing. Also confirm on your dashboard at teams.igem.org/6462 which Village we are registered under. Teams in the Software and AI Village cannot win Best Software Tool at all, so if that is us, this whole prize track is off the table and we should know that now, not in October.

How the pieces fit together



Read it left to right. The kinetics model and the structural scorer together are the substance of the Model prize entry. The structural scorer's output also feeds the future software tool, because that tool is just both models wrapped in something clickable. The software tool box is dashed because it does not exist yet. That is deliberate.

Who owns what

Person	Workstream	One line mandate
Archer & Aniu	Kinetics model	Build and validate the maths that predicts how fast and how strongly our construct signals, from cleavage to readout.
Aaran	Structural scorer	Estimate, from protein structure, how likely a given tag placement actually locks the hairpin shut. Feed that number into the kinetics model.
Orlando	Wiki (now)	Build the site skeleton, including placeholder Model and Software pages at the correct standard URLs, so content has somewhere to land.
Orlando & Aaran	Software tool (later)	Once both models above have a working version one, wrap them into a tool other teams can run. Spec only, for now.

Workstream 1: the kinetics model (Archer and Aniu)

The maths, as a starting skeleton

This is not the finished model. It is the skeleton you build on. Four linked stages, each one a rate.

Stage 1. Cleavage. MMP14 cuts the SGRSENIRTA linker. In the regime where substrate is far below K_M, the standard low substrate linearisation applies:

$$\frac{d[S]}{dt} = -\frac{k_{cat}[E]}{K_M}[S], \qquad [I] \text{ accumulates as } [S] \text{ depletes}$$

where $[S]$ is intact construct, $[E]$ is active enzyme concentration, and $[I]$ is cleaved but not yet structurally relaxed.

Stage 2. Conformational release. Cleavage does not instantly open the hairpin. The two protein fragments have to physically separate before the loop relaxes:

$$\frac{d[I]}{dt} = \frac{k_{cat}[E]}{K_M}[S] - k_{open}[I], \qquad \frac{d[S^*]}{dt} = k_{open}[I]$$

k_{open} is the rate nobody has measured yet. It is exactly what the structural scorer in workstream 2 estimates. This is the join between the two workstreams, treat it as a single shared parameter, not two separate guesses.

Stage 3. Strand displacement. The open construct S^* exposes a toehold and reacts with a downstream reporter gate G:

$$\frac{d[P]}{dt} = k_{eff}[S^*][G]$$

k_{eff} depends on toehold length and is an experimentally characterised curve, not a free parameter, see the parameter table below.

Stage 4, optional. Amplification. If we are feeding this into a Cas12a readout, collateral cleavage of a reporter follows saturating kinetics:

$$\frac{d[\text{signal}]}{dt} = \frac{k_{cat,Cas}[\text{Cas:gRNA:target}][R]}{K_{M,Cas} + [R]}$$

Parameters to pull from the literature first, not guess

Parameter	Value or range	Source
k_{cat}/K_M, MMP family on PEG cleavable substrates	approx 2×10^5 $M^{-1}s^{-1}$	Kwong et al, PNAS 2015. pnas.org/doi/10.1073/pnas.1506925112
SGRSENIRTA selectivity	79% MMP14, 4% MMP2, 9% MMP9	Immunomodulatory MMP review. pmc.ncbi.nlm.nih.gov/articles/PMC9800621
k_{eff}, toehold exchange, 0 to 7 nt toehold	roughly 1.4 to 1×10^6 $M^{-1}s^{-1}$	Zhang and Winfree, JACS 2009. pubmed.ncbi.nlm.nih.gov/19894722
k_{open}	unknown, our job to estimate	Workstream 2 output.

Task list, in order

- ☐ Learn how RK4 (fixed step, fourth order Runge-Kutta) and Runge-Kutta-Fehlberg (adaptive step, RKF45) integrators actually work, not just how to call a library function. Practice on a toy ODE first, for example exponential decay or a simple predator-prey pair, and check your numerical answer against the known analytic solution before trusting either method on the real model. This is not busywork, Visual DSD's own deterministic simulator runs on Runge-Kutta-Fehlberg, so you are learning the actual mechanism behind a tool referenced elsewhere in this document.
- ☐ Write the four stage system above as a working ODE in Python or Mathematica. Deliverable: a script that takes k_{cat}/K_M, k_{open}, k_{eff}, and enzyme concentration as inputs and plots signal over time.
- ☐ Pull the literature values in the table above into the model as defaults. Confirm each number against the source, do not trust this table blindly.
- ☐ Run a parameter sweep. For each of the four rate constants, vary it across a realistic range and record how it changes two outputs: time to half maximum signal, and the minimum enzyme concentration that gives a detectable signal within a fixed assay time.
- ☐ Run a proper sensitivity analysis, Sobol indices or an equivalent variance based method, not just eyeballing the sweep. Output: a ranked list of which parameter actually controls detection limit and speed.
- ☐ Once Aaran has a k_{open} estimate from workstream 2, plug it in and rerun. This is the point where the model becomes predictive rather than illustrative, which is the entire difference between a Best Model entry and a paragraph that mentions Michaelis-Menten.
- ☐ Once wet lab data exists, fit the model to it and report the comparison plot. This is the validation step the judges explicitly ask for, it is not optional.
- ☐ Write the model wiki page narrative. State the equations, state the parameters, state what the model predicted before the wet lab ran, then show how it compared.

Deliverables, Archer and Aniu

Item	Lives where
Working ODE model, code and plots	Team GitLab repo
Sensitivity analysis writeup	Model wiki page
Literature parameter table, checked against sources	Model wiki page
Model versus wet lab validation plot, joint with Aaran	Model wiki page

Workstream 2: the structural feasibility scorer (Aaran)

Be honest about what this is

Training a neural network from scratch on five labelled examples is still a bad idea. No amount of personal preference changes that, you would overfit and the result would be noise dressed up as a model. But there is a real, defensible way to make this a neural network problem rather than a physics problem: do not train from scratch, do what you already do with LLMs at Uni Customs, take a model pretrained on a huge corpus and adapt it. Protein language models, ESM2 from Meta being the standard one, are transformers trained on hundreds of millions of real protein sequences. They already encode structural and evolutionary signal about flexibility, solvent exposure, and local geometry, the exact things a hand built physics score is trying to approximate badly from five examples. Pulling embeddings from a pretrained model and calibrating a small layer on top is closer to your actual skill set than fitting a polymer physics curve by hand, and it is a stronger story for the wiki: a team that recognised data scarcity and reached for transfer learning, instead of either overfitting a deep net or hand waving with physics, looks more credible to a judge, not less. A 2025 *Scientific Reports* paper makes exactly this point, medium sized protein language models transfer well precisely when the downstream dataset is small, bigger is not automatically better here.

The approach

Two layers, both genuinely neural. Layer one, structure: AlphaFold3 or ESMFold predicts the 3D structure from sequence. Both are large transformer based models, so the moment a structure is predicted, deep learning is already running, that part of the pipeline was never "not ML". Layer two, scoring: for each candidate pair of tag insertion sites, extract per residue embeddings from a pretrained ESM2 model at those positions, the same extract embeddings from a frozen model move already used with LLMs elsewhere on this CV. Combine those embeddings with the geometric features (through space distance, backbone path length, solvent accessibility) inside a small, heavily regularised model: logistic regression or a one layer network with strong L2 penalty, not a deep net. The pretrained representation does the heavy lifting, the trainable layer on top only needs enough capacity to combine features, not enough to memorise five data points.

Task list, in order

- ☐ Get a structure for the scaffold protein. Use a solved PDB structure if one exists for our construct, otherwise predict one with AlphaFold3 or ESMFold. Either way, note for the wiki that this step is itself a transformer doing the work.
- ☐ Pull per residue ESM2 embeddings at each candidate tag insertion site. The `facebookresearch/esm` repository ships a ready made extraction script, there is no need to write a custom forward pass.
- ☐ Also compute the geometric features: through space distance between the two residues, path length along the backbone, and a flexibility proxy such as predicted B factor or pLDDT. These feed in alongside the embeddings, physics is not abandoned, it is just not doing all the work anymore.
- ☐ Train a small, heavily regularised model, logistic regression or a one layer network with strong L2 penalty, on the combined embeddings and geometric features. Calibrate it against the handful of real precedent geometries that exist in the literature: the bioorthogonal protein DNA conjugation force spectroscopy paper and the Merkx group's protease cleavable DNA lock work, both of which report real measured or designed inter point distances.
- ☐ Package this as a script: input a structure file and two residue numbers, output a k_{open} band (low, medium, or high, not a single false precision number) and a one paragraph justification.
- ☐ Hand the output to Archer and Aniu for workstream 1. Write up why transfer learning was the right call here, citing the 2025 transfer learning paper, as a short section for the model wiki page.

Deliverables, Aaran

Item	Lives where
Structural scorer script, ESM2 embeddings plus calibrated layer	Team GitLab repo, model wiki page
Transfer learning rationale writeup, with citation	Model wiki page
k_{open} estimate, handed to Archer and Aniu	Recorded in repo, used in workstream 1
Model versus wet lab validation plot, joint with Archer and Aniu	Model wiki page

Workstream 3: the wiki (Orlando)

- ☐ Read the wiki requirements page in full: teams.igem.org/6462/deliverables/wiki/requirements.
- ☐ Look at two strong example wikis for structure ideas, not content to copy: 2023.igem.wiki/mcgill and 2025.igem.wiki/greatbay-scie.
- ☐ Draft a full site map: every page we need, including the standard pages required for medals and special prizes (Model, Software, Contribution, Engineering, Human Practices).
- ☐ Build the site skeleton on iGEM's GitLab now, with placeholder Model and Software pages already sitting at the correct standard URLs, even if empty. The URL must exist and must not move later.
- ☐ Get the GitLab CI build pipeline working end to end before anyone needs to push real content. Test this early, not the week of the freeze.

Deliverables, Orlando

Item	Lives where
Full site map document	Team GitLab repo
Wiki site skeleton with all standard page URLs live	2026.igem.wiki, confirm slug on dashboard
GitLab CI pipeline, tested and working	Team GitLab repo

Workstream 4: the software tool, later (Orlando and Aaran)

Once workstream 1 and workstream 2 each have a working first version, this workstream starts. The plan, so it exists on paper now:

- ☐ A small tool that takes a candidate construct as input: target protease, cleavage motif, candidate tag positions, toehold length, and returns a predicted signal curve plus a design recommendation. It is workstream 1 and workstream 2, wrapped.
- ☐ Shape it like Lambert iGEM 2025's CASPER tool: a web app for anyone to click through, plus a command line and Python package version so it is genuinely useful to other teams, not just a demo. Reference: 2025.igem.wiki/lambert-ga/software.
- ☐ Host the source on iGEM's GitLab software tools repository, the 2026 path will follow the pattern gitlab.igem.org/2024/software-tools, confirm the exact 2026 path on the dashboard.
- ☐ Decision point, to be made as a team, not assumed: if there is enough time left once the model is solid, Orlando and Aaran start building together. If not, we drop this prize track entirely and put all remaining effort into the Model entry and the wiki. A half built tool scores worse on the usability and documentation questions than no tool at all, so this is a real either or, not a stretch goal to half attempt.

Deliverables, Orlando and Aaran

Item	Lives where
Go or no go decision on the software tool, written down	Recorded in team GitLab repo
If go: working web app, CLI, and package, hosted before GitLab freeze	gitlab.igem.org software tools repo

Resources

- iGEM judging handbook: static.igem.org/websites/competition/2025/judging/2025judgehandbook.pdf
- iGEM special prizes overview: competition.igem.org/judging/awards/special
- Wiki requirements for our team: teams.igem.org/6462/deliverables/wiki/requirements
- Example wiki, McGill 2023, Grand Prize winner: 2023.igem.wiki/mcgill
- Example wiki, GreatBay SCIE 2025, heavy computational model page: 2025.igem.wiki/greatbay-scie/model
- Example software tool, Lambert 2025 CASPER: 2025.igem.wiki/lambert-ga/software
- Kwong et al, multicompartment activity based biomarker model, PNAS 2015: pnas.org
- Zhang and Winfree, toehold exchange kinetics, JACS 2009: pubmed.ncbi.nlm.nih.gov/19894722
- SGRSENIRTA selectivity data: pmc.ncbi.nlm.nih.gov/articles/PMC9800621
- NUPACK, nucleic acid design and analysis suite: nupack.org
- ESM2, pretrained protein language model, Meta: github.com/facebookresearch/esm
- Why pretrained embeddings beat training from scratch on small datasets, Scientific Reports 2025: nature.com/articles/s41598-025-05674-x
- Multistrand, nucleic acid kinetics simulator: github.com/DNA-and-Natural-Algorithms-Group/multistrand
- Visual DSD and CRN engine, strand displacement circuit design: github.com/microsoft/crn-engine